

VALTIMO KK, Finland

WMO: 027980

Lat: 63.667N

Lon: 28.829E

Elev: 115

StdP: 99.95

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.6	(c) -27.4	(d) -33.8	(e) 0.2	(f) -30.6	(g) -30.1	(h) 0.2	(i) -27.3	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.2	27.0	18.3	24.9	17.4	23.0	16.5	20.0	24.6	18.7	23.3	17.6	21.4	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	13.4	21.7	17.1	12.4	20.5	15.8	11.4	19.4	57.7	24.8	53.6	23.3	49.9	21.2	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) N/A	(b) N/A	(c) N/A	(e) -33.0	(f) 28.6	(g) 4.5	(h) 3.2	(i) -36.3	(j) 30.9	(k) -38.9	(l) 32.8	(m) -41.5	(n) 34.6	(o) -44.7	(p) 36.9		
(4) N/A	(5) N/A	(6) N/A	DB	WB	-32.8	21.4	4.4	2.1	-36.0	22.9	-38.5	24.2	-41.0	25.4	-44.2	26.9

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.0	-11.2	-8.4	-4.8	1.6	8.9	13.1	16.6	14.3	9.6	2.9	-1.4	-6.2
	DBStd	10.51	8.42	7.79	5.65	3.97	4.62	3.98	3.59	3.33	3.47	3.60	4.50	7.16
	HDD10.0	3096	656	516	459	253	77	14	1	5	49	219	342	502
	HDD18.3	5643	914	749	718	503	294	164	77	131	262	477	592	761
	CDD10.0	531	0	0	0	1	42	107	205	139	38	1	0	0
	CDD18.3	36	0	0	0	0	1	6	23	6	0	0	0	0
	CDH23.3	367	0	0	0	0	34	72	204	57	0	0	0	0
	CDH26.7	66	0	0	0	0	2	11	43	9	0	0	0	0
Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Precipitation	PrecAvg	657	47	38	33	34	51	66	86	71	58	60	59	56
	PrecMax	894	88	80	68	60	103	134	161	165	110	108	128	113
	PrecMin	443	13	4	6	9	12	20	33	17	13	12	12	17
	PrecStd	96	18	19	14	14	25	29	37	32	25	26	25	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.1	4.5	8.0	17.6	26.5	28.1	29.9	27.9	20.5	12.4	8.5	4.2
		MCWB	1.5	2.8	3.8	9.6	14.8	18.2	20.8	19.9	15.6	10.8	7.5	3.7
	2%	DB	1.1	2.7	5.0	12.9	23.9	25.7	27.7	24.7	18.2	10.6	6.5	2.6
		MCWB	0.7	1.6	1.9	7.5	14.3	17.3	19.4	17.6	14.3	9.3	5.9	1.9
	5%	DB	0.3	1.4	3.3	10.3	20.7	22.9	25.5	22.5	16.3	9.3	5.0	1.6
		MCWB	0.0	0.8	1.3	5.4	13.0	15.8	18.5	16.7	13.4	8.2	4.6	1.0
	10%	DB	-0.4	0.5	2.1	8.1	17.6	20.7	23.6	20.5	14.9	8.0	3.7	0.8
		MCWB	-0.6	0.1	0.6	4.3	11.0	14.5	17.5	15.9	12.8	7.1	3.3	0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	3.1	4.1	10.3	16.3	19.7	22.0	21.1	16.3	11.2	7.9	4.0
		MCDB	2.2	4.2	6.7	15.5	23.5	24.1	26.9	26.8	18.8	11.9	8.5	4.2
	2%	WB	0.8	1.9	2.6	8.1	15.1	18.3	20.7	18.7	15.2	9.7	6.0	2.1
		MCDB	1.0	2.5	4.4	11.9	21.4	23.8	25.4	23.3	17.2	10.3	6.4	2.4
	5%	WB	0.1	0.9	1.7	5.9	13.6	16.6	19.6	17.5	14.0	8.4	4.7	1.2
		MCDB	0.3	1.3	2.9	9.3	19.8	21.6	24.2	21.0	15.8	9.1	5.0	1.5
	10%	WB	-0.7	0.2	0.9	4.7	11.7	15.2	18.4	16.5	13.0	7.1	3.3	0.5
		MCDB	-0.4	0.6	1.8	7.3	16.9	19.4	22.5	19.6	14.6	8.0	3.6	0.8
Mean Daily Temperature Range	5% DB	MDBR	6.2	7.1	9.2	9.3	10.3	9.5	9.2	8.7	7.1	4.9	3.7	5.1
		MCDBR	3.7	4.5	8.0	13.7	15.6	13.3	12.4	12.3	9.4	6.3	4.1	3.6
		MCWBR	3.8	4.1	6.0	8.2	7.3	6.0	5.3	6.0	6.0	5.1	3.9	3.4
	5% WB	MCDBR	3.5	4.4	6.5	11.7	14.0	11.1	11.0	10.0	8.3	5.9	4.1	3.4
		MCWBR	3.6	4.2	5.2	7.4	7.2	5.4	5.4	5.6	5.6	5.3	3.9	3.4
Clear-Sky Solar Irradiance	taub	0.220	0.281	0.309	0.347	0.348	0.346	0.371	0.352	0.329	0.307	N/A	0.209	
	taud	1.976	2.104	2.155	2.266	2.447	2.478	2.478	2.423	2.462	2.516	2.393	N/A	1.967
	Ebn at Noon	364	626	771	828	866	875	836	822	771	619	N/A	164	
	Edh at Noon	30	65	93	105	97	96	99	87	68	50	N/A	13	
All-Sky Solar Radiation	RadAvg	0.17	0.72	2.17	3.68	4.69	5.23	4.93	3.74	2.05	0.76	0.18	0.07	
	RadStd	0.02	0.16	0.24	0.41	0.52	0.33	0.52	0.41	0.23	0.13	0.04	0.01	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)		+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page